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FiRe: Fine-grained Multimodal Reasoning for Enhanced Image Generation

Requirement-Level Check-Refine Cycles and Per-Step RL Rewards Fix the Fix-One-Break-Another Problem in Compositional T2I

By Yongjin Kim, Yoonjin Oh, Yerin Kim, Hyomin Kim, Jeeyoung Yun, Yujung Heo, Minjun Kim, Sungwoong Kim (KU-AGI)

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Text-to-image models fail on compositional prompts because whole-image feedback cannot isolate which individual visual requirement is violated—so fixing one element inadvertently breaks another. FiRe decomposes prompts into atomic visual requirements, checks each one independently using a multimodal LLM, and applies *local* refinements to only the failing regions. FiRe-GRPO, the accompanying reinforcement learning variant, assigns a per-step reward to each check-refine action, enabling the model to learn when and how to refine rather than merely whether the final image is correct. On compositional benchmarks including T2I-CompBench and GenAI-Bench, FiRe achieves consistent improvement over T2I baselines, with the largest gains on prompts specifying three or more simultaneous visual constraints.

AT A GLANCE

Problem: Compositional T2I prompts impose multiple simultaneous visual constraints; whole-image feedback cannot localize failures or guide targeted corrections.

Why it matters: State-of-the-art T2I models still fail on multi-constraint prompts despite improved language understanding; the bottleneck is the feedback signal, not the generator.

Method: Decompose the prompt into visual requirements; check each with an MLLM; refine locally; train with FiRe-GRPO assigning step-level rewards to each check-refine action.

Result: Consistent improvement on compositional benchmarks, especially on prompts with ≥ 3 simultaneous constraints.

Evidence: T2I-CompBench, GenAI-Bench; ablations isolate local vs. global refinement and step-level vs. terminal-only reward.

The Big Picture

Language-guided image generation has improved dramatically, yet compositional prompts—requests that simultaneously specify object counts, attributes, spatial relationships, and interactions—remain a known failure mode. A prompt such as "a red cup to the left of a blue vase on a wooden table" imposes three simultaneous visual constraints; any failure in the generated image requires identifying *which* constraint is violated before attempting a correction.

Existing reasoning-augmented approaches attempt to fix this by elaborating the prompt before generation or by scoring the whole image after generation. Neither approach can localize the failure to a specific requirement or avoid disrupting already-satisfied constraints during refinement.

FiRe reframes image generation as a sequential decision process where each step either checks one requirement or applies one local refinement, and learns via reinforcement learning to execute this process efficiently.

Why Existing Approaches Fall Short

Prompt augmentation (chain-of-thought elaboration): Improves average image quality but generates the same elaborated prompt for every seed; it cannot recover from specific failures in a generated image.

Whole-image feedback (DALL-E Reflect, ImageReward): Produces a scalar or sentence-level critique. This is too coarse to guide targeted corrections and does not indicate which region or property to change.

Terminal-only RL for T2I: Assigning reward only to the final image creates a sparse signal that is difficult to optimize and does not credit or penalize intermediate refinement decisions.

Core Mechanism

Step 1 — Requirement decomposition. An MLLM parses the prompt into a list of atomic visual requirements $\mathcal{R} = \{r_1, \ldots, r_n\}$, where each r_i is a checkable property (e.g., "red cup present", "cup to the left of the vase").

Step 2 — Initial generation. A base T2I model generates I_0 from the prompt.

Step 3 — Fine-grained check. For each unsatisfied requirement r_i, the MLLM answers "Is r_i satisfied in I_{t-1}?" with a binary verdict and a natural-language diagnosis specifying the region and nature of the failure.

Step 4 — Local refinement. The least-satisfied requirement $i^* = \arg\min_i \mathcal{V}_i(I_{t-1})$ triggers an inpainting or latent edit confined to the region described in the diagnosis, yielding I_t. Unaffected regions are left unchanged.

Step 5 — Iteration. Steps 3–4 repeat until all requirements are satisfied or the step budget T is exhausted.

Technical Formulation

Let $\mathcal{V}_i(I) \in [0,1]$ denote the MLLM's estimated probability that requirement r_i is satisfied in image I. The step-level reward at refinement step t (targeting requirement i^*) is:

$$\rho_t = \mathcal{V}_{i^*}(I_t) - \mathcal{V}_{i^*}(I_{t-1})$$

The FiRe-GRPO policy gradient objective is:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}\left[\sum_{t=1}^{T} \rho_t \cdot \log \pi_\theta(\delta_t \mid I_{t-1}, r_{i^*})\right] - \beta \cdot D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})$$

where δ_t is the refinement action (diagnosis + edit), π_θ is the MLLM policy being trained, and π_{ref} is a frozen reference policy. The KL term prevents the policy from diverging far from a well-behaved prior and controls mode collapse.

The combined requirement satisfaction score at the end of generation is:

$$\text{Score}(I_T) = \frac{1}{n} \sum_{i=1}^{n} \mathcal{V}_i(I_T)$$

This is the final metric reported on benchmarks.

Learning or Inference Procedure

Training:

1. Sample a compositional prompt; decompose into requirements.
2. Generate I_0 with the base T2I model.
3. Run the check-refine cycle for T steps; collect $\{\rho_t\}$.
4. Update the MLLM backbone with FiRe-GRPO.
5. The 300-step checkpoint (FiRe-300Step) is used for all evaluations.

Inference:

1. Parse the prompt; generate I_0.
2. Iterate: check all requirements, identify the most violated, apply a local refinement.
3. Return I_T after the budget is exhausted or all requirements are satisfied.

What the Guarantee Says

The step-level reward ρ_t provides a per-requirement improvement signal. Under mild assumptions on the MLLM's calibration (i.e., $\mathcal{V}_i$ monotonically correlates with human satisfaction judgments), the GRPO objective with the KL regularizer converges to a policy that maximizes expected total requirement satisfaction while remaining close to the reference policy. This ensures that FiRe does not sacrifice the reference model's general image quality in pursuit of compositional accuracy.

Experimental Findings

- **T2I-CompBench:** FiRe improves attribute binding and spatial relationship accuracy over all baselines; largest gains on prompts with ≥ 3 simultaneous constraints.
- **GenAI-Bench:** Consistent improvement across compositional categories (color, count, spatial, interaction).
- **Fix-one-break-another:** Global refinement baselines show improvement in the targeted attribute but regression in other attributes; FiRe's local edits preserve already-satisfied requirements.
- **FiRe-GRPO vs. supervised fine-tuning:** Step-level RL converges faster and produces better refinement policies than training with supervised labels for each step.

Ablations and Interpretation

Global vs. local refinement: Applying refinement to the whole image rather than the failing region degrades already-satisfied requirements by an average of 12 percentage points on T2I-CompBench; local refinement avoids this regression entirely.

Step-level vs. terminal-only reward: Terminal-only reward (only $\mathcal{V}_{i^*}(I_T)$ at the final step) trains 40% slower and achieves 8 percentage points lower compositional accuracy; step-level rewards provide a denser credit assignment signal.

Number of refinement steps T: Performance improves with T up to 3–4 steps and plateaus beyond; additional steps add compute without further gains on typical 3-constraint prompts.

Without requirement decomposition: Using a single holistic verdict instead of per-requirement checks degrades performance to below the whole-image feedback baseline, confirming that the decomposition is the key structural contribution.

Reference

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